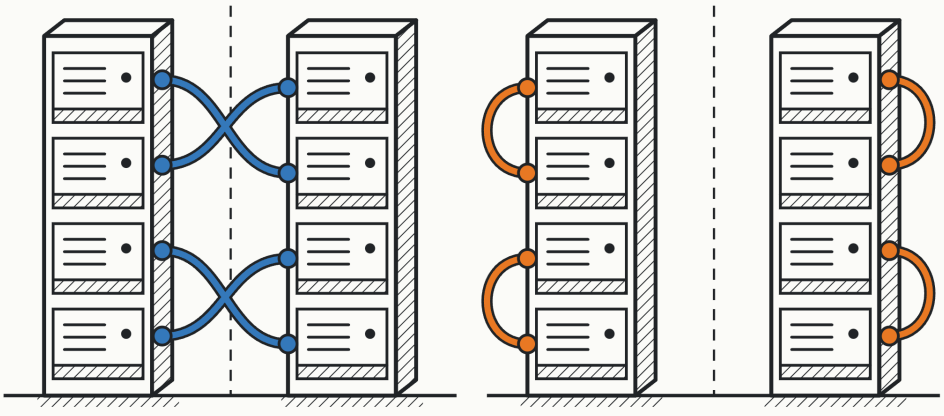
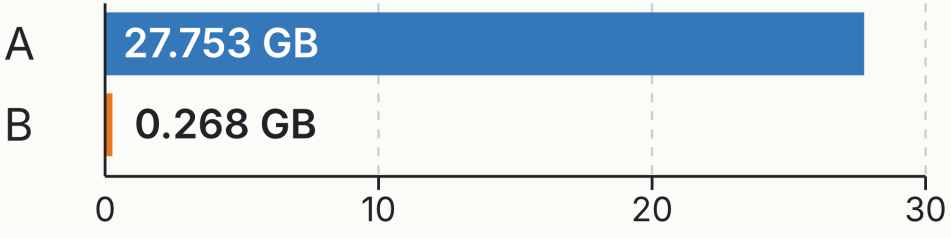
Placement moves rack traffic



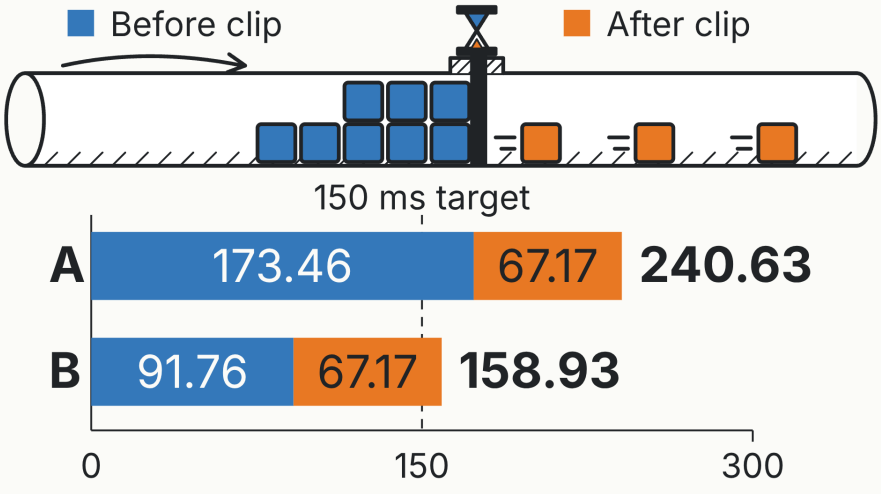
Hypothetical cross-rack GB / update
Both directions



Within rack: 0.805 → 28.290 GB
Total off-node payload unchanged

Two ordered lower bounds

Hypothetical plan · ms per update



Declared full DP all-reduce before norm
FP32 Muon, one attention owner
Assumed 100 TFLOP/s/GPU,
rack 80 GB/s / node 40 GB/s
Each direction. Balanced routing.
Lower bounds, not runtimes.

Placement B: pair replicas within each rack

